

eyemR-Vis: Using Bi-Directional Gaze Behavioural Cues to Improve Mixed Reality Remote Collaboration

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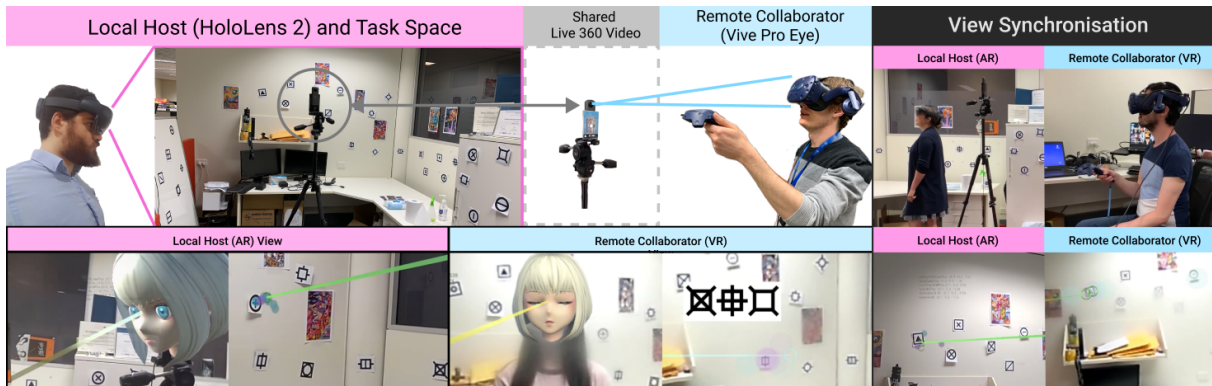


Figure 1: The eyemR-Vis prototype system, showing an AR user (HoloLens2) sharing gaze cues with a VR user (HTC Vive Pro Eye)

ABSTRACT

Gaze is one of the most important communication cues in face-to-face collaboration. However, in remote collaboration, sharing dynamic gaze information is more difficult. In this research, we investigate how sharing gaze behavioural cues can improve remote collaboration in a Mixed Reality (MR) environment. To do this, we developed eyemR-Vis, a 360 panoramic Mixed Reality remote collaboration system that shows gaze behavioural cues as bi-directional spatial virtual visualisations shared between a local host and a remote collaborator. Preliminary results from an exploratory study indicate that using virtual cues to visualise gaze behaviour has the potential to increase co-presence, improve gaze awareness, encourage collaboration, and is inclined to be less physically demanding or mentally distracting.

CCS CONCEPTS

• **Human-centered computing** → **Visualization design and evaluation methods**; **Mixed / augmented reality**; **Interactive systems and tools**.

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KEYWORDS

Gaze Visualisation, Human-Computer Interaction, Mixed Reality Remote Collaboration, CSCW

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1 INTRODUCTION

Gaze cues in face-to-face communication can be used to exchange complex social signals [4], predict another person's intention [2], understand the desire to communicate [13] or seek for approval [8]. Eye movements such as fixations, saccades, and blinks are common gaze behaviours [30] and for successful communication people typically exchange a range of different gaze signals. Similarly, there is also a gaze-speech relationship between the gaze direction of two conversation partners [30] and knowing what they are referring to is critical for task coordination during communication.

Researchers have explored the use of gaze cues compared to other visual communication cues for both co-located [23][3] and remote task spaces [15][1] through traditional [32][7] and immersive technologies (Augmented Reality (AR) and Virtual Reality (VR)) [26][18]. In many Mixed Reality (MR) collaboration studies[9], gaze is often uni-directional [20][11] or considered as a combined research subject [27][1]. In AR and VR, gaze cues are often illustrated as a simple marker representing the current fixation location [15][18] or as a

ray converging on a surface of the target [22][1], yet visualisation of gaze behaviours has not been explored actively in immersive interfaces.

In traditional 2D displays, different gaze behaviours have been visualized in varied ways[7]. However, traditional displays have limited screen space and are poor at conveying spatial cues, so using a variety of gaze visual information possibly cause user "confusion" and "distraction" [32][24]. Therefore, findings from desktop systems may not be fully relevant to immersive experiences. This identifies a need for further investigation of visualisation as well as sharing of gaze behaviours in collaborative MR environments to naturally represent the rich gaze behavioural cues that exist between collaborators.

To investigate the impact of visualisation of gaze behaviours in immersive applications, we developed *eyemR-Vis* (Fig. 1). This is a prototype MR collaboration system that depicts bi-directional gaze behavioural information with natural and subtle visualisations during remote task collaboration. While supporting explicit gaze visualisation, the system aims to reduce distracting characteristics such as rigid gaze visual feedback that has been used in previous research. We expect our investigation to open up more opportunities to explore how visualising and sharing dynamic gaze behaviours can lead to better practices in MR remote task collaboration.

Our research gives rise to the following contributions:

- The first system to incorporate bi-directional gaze behavioural cues in 360 panorama based MR remote collaboration.
- A pilot study that compares bi-directional gaze to commonly used uni-directional gaze against gaze behavioural cues.
- The first collaborative MR system that changes gaze visualisation styles depending on user gaze behaviour.

2 RELATED WORK

In collaborative tasks, gaze is one of the crucial communication cues for enhancing coordination strategies [32]. Eye fixation, for instance, can serve as a fast and precise pointer [12], confirm and clarify the object of interest [7], or provide a reference that simplifies linguistically complex objects with deictic references (e.g. "this" or "here") [21][6]. Also, mutual gaze awareness can lead to joint attention, resulting in reaching successful common ground in collaborative decision-making [31]. Sharing gaze can not only improve the awareness of collaborator's focus [18][17] but also enhance the feeling of co-presence [11][1] in a remote setting.

In previous studies on gaze using desktop displays, researchers have used different visualisations such as a cursor, spotlight, scan-path, trail or heatmap to represent real-time gaze behaviours for different purposes [3][32][24][7]. This is an effective aid to partition task visual space [3] or for interpreting a partner's intention [24], but these different gaze visualisations can also add mental distraction [32][18].

Most previous research has used gaze to interact with desktop screens [6][5][7] or 2D displays [32][24], and the exploration of different visual representations for gaze behavioural cues such as saccades, tracking movement, or fixations in MR remote collaboration environment remains limited. The most common way to visualise gaze in MR systems is to use a simple cursor (e.g. a solid circle or a crosshair) to represent the current gaze point tracked

in real-time [18][11][12]. Another widely used method is a virtual gaze ray [22][10], or even combining a view frustum with a gaze ray in shared AR and VR task space [26][1]. However, prior research visualised gaze in a simple fixed representation without changing gaze visualization depending on the user's gaze behaviour. For example, fixation on an object may imply a user's interest compared to rapid gaze browsing, or two users looking at the same point may indicate mutual awareness of each other's focus. Not showing different gaze representations may hinder the exchange of precise gaze cues for communication in a spatially complex task when collaborators aren't physically together.

In our work, we explore how subtle yet natural bi-directional gaze visualisation cues can represent four gaze behaviours in collaborative MR interfaces, and thus impact user collaboration. While there is prior work that has used live 360 panorama for MR collaboration [19][14][28][29], to the best of our knowledge, our research is the first example of integrating gaze cues with live 360 panorama based MR collaboration and visualising different gaze behaviours. Through our research, we attempt to fill the gap between a relatively mature body of research on MR remote collaboration and the lack of research on the representation of dynamic gaze cues during MR collaborative tasks.

3 SYSTEM DESIGN

eyemR-Vis is a live 360 panoramic video-based Mixed Reality remote collaboration system that visualises various gaze behavioural cues in a bi-directional virtual interface. The local host (the user in the physical task space) shares and receives verbal and gaze communication cues using an AR headset to and from a remote collaborator in VR, while also sharing a live 360 panoramic video of his/her surroundings. In Fig. 1, the local host is equipped with a see-through AR head-mounted display (HMD) with an eye-tracker and a 360 panoramic video capturing device placed at the centre of the task space, while the remote collaborator wears a VR HMD with an eye-tracker to view the shared live scene. Collaborators can exchange their gaze cues via eye-trackers built into their HMDs in a bi-directional manner. The system is designed to facilitate pair tasks between users who are not physically co-located in the task space by explicitly adding visualisations of gaze behaviour cues from both sides into the collaborative MR interface. In this way, both the AR and VR users know exactly where their collaborator is looking and the type of gaze behaviour they are exhibiting.

3.1 360 Mixed Reality View

To mimic the task space as close as possible to the local host's environment, providing an independent omni-directional natural view for the remote collaborator is essential. To achieve this we position a live 360 camera in the centre of the task space which captures and shares the environment and the local user's actions during the collaboration (Fig. 2 - a1). The shared panoramic video allows the remote collaborator in VR to freely determine his/her own view and probe the surroundings independently from the local host's control. In the meantime, the system also lets the VR user zoom in (Fig. 2 - a2) by toggling a second camera view that is placed forward from where the centre camera faces. This provides a close

up omni-directional viewpoint to check the situated objects of interests to compensate for the immobility that the remote collaborator experiences. The overall effect is that the VR user feels that s/he is standing in the same real environment as the local AR user while being able to have completely independent control of their view.

3.2 Gaze Cues Visualisation and Interaction

Cursors and virtual gaze rays are the most widely used view direction cues for MR collaborations [10][22][26][18][11]. In our system, we use a combination of a virtual gaze ray which starts from collaborator's HMD and a cursor which is placed at where the ray and a physical surface intersect to clearly indicate the target of interest [16]. In addition, we also add a virtual avatar head from where the ray starts. From the local host view, the remote user's avatar head sits at where the 360 camera is placed (Fig. 1 - row 2 left), while the remote collaborator sees it superimposed on top of the local host's HMD in the 360 live video (Fig. 1 - row 2 middle). For the users, their own gaze is presented only as a cursor to reduce distraction [32] and mental load [18].

To synchronise and amplify gaze behavioural cues between collaborators, we have designed four different visualisations styles. 1) Browsing state (Fig. 2 - a1 and a2): defined as active eye movements such as saccades, blink, or shift in purple (own gaze) and blue (partner's gaze); 2) Focus state (Fig. 2 - b1): when the user's gaze dwells on the same point for 500 ms or more [25]. When this happens that gaze visualisation turns into a yellow colour; 3) Mutual state (Fig. 2 - b2): we change both collaborators gaze visualisations green to indicate gaze overlap (joint gaze awareness); 4) Fixated circle-map (Fig. 2 - b3): we use semi-transparent coloured circles (8% opacity) to map the latest ten focus locations for each collaborator. This visualisation is a variation of the traditional heatmap design where a colour coding (e.g. green-yellow-red) is used permanently to represent dwell duration. To avoid distraction we reduce the colour to one and use the opacity to represent gaze dwell duration (increasing 8% every 500ms at a max of 80% opacity). We also limit the numbers of circles visualising to only the latest ten per collaborator to avoid full occlusion triggered by accumulated layers of semi-transparent circles on top of each other over time as this may discourage collaborators to look at the same point together [7].

3.3 User Interviews and Design Iteration

Over the process of design and development of the prototype, we presented it to several sample users and got early feedback from interviews so we could do a design iteration. We discovered that the amount of displayed gaze information can possibly increase user distraction. Therefore, we improved the gaze interfaces in the following ways: 1) Removed the real-time trajectory gaze trail and replaced it with the latest ten fixated circle-map; 2) Enabled the choice of a stacked circle-map (circles accumulate on top of each other) or in a single layer form (circles won't be generated if the current location has one), based on task properties; 3) Added an option to enable/disable the user's personal gaze cursor (own gaze indicator); 4) Included a mutual state colour indicator (as described in Section 3.2) which unifies collaborators respective gaze colours (blue and purple) into green when they overlap (joint gaze).

4 IMPLEMENTATION

The prototype system is built with Ricoh Theta V ¹ 360 panorama camera (supports 1080p and 4k resolution) to capture the task space of the local environment. The local host is equipped with the HoloLens2 AR HMD ² to map the local task environment as well as using the Vuforia tracker (v9.3.3) ³ and a visual marker positioned on the 360 camera to align both the local and remote coordinate systems. The remote collaborator wears a Vive Pro Eye VR HMD ⁴ running on a PC (Intel Core i7-6700 3.40GHz CPU, 16GB DDR4 RAM, NVIDIA GeForce 950 GPU) with the Windows 10 Operating System. To capture eye gaze data we use the HoloLens2 and Vive Pro Eye built-in eye trackers. Participants need to successfully calibrate their eyes before setting up the entire system. The MRTK 2.4.0 ⁵ and SRanipal SDK ⁶ are used to turn data into visualisations respectively, while the SteamVR Unity plugin (v2.6.1) ⁷ is used to support PC-Controller-HMD interactions. The VR side works as the server for easy monitoring purpose while AR joins as a client when the server is running. Avatars are picked from the prefabs SRanipal example uses and are programmed in a 1:1 real-life ratio to constantly sync with where collaborators are facing. Finally, a Vive controller is used for changing view options: users can press down the trigger button to zoom in to have a closer 360 omni-view, while the grip button is used for the remote collaborators to toggle reference on/off for the asymmetric guiding task. Both the AR and VR experiences are developed using the Unity game engine (v 2019.3.15) ⁸.

5 EXPLORATORY USER STUDY

We recruited eight participants in pairs (2 female and 6 male; aged between 20 - 32) to conduct a preliminary exploratory study. The user feedback consisted of both qualitative interviews as well as quantitative ratings and preference ranking on questionnaires.

5.1 Study Design and Procedure

As an exploratory study, we collected feedback from participants after they tried different combinations of gaze visualisation options while performing collaborative tasks. Although our main focus was to investigate the gaze behaviour, we also wanted to test whether showing the user's personal gaze (own gaze indicator) is useful or not, as prior works reported conflicting results [18]. Therefore, we let paired participants test the prototype system under four conditions that are combinations of two factors as shown in Table 1: gaze behaviour on/off and own gaze visualisation on/off (i.e. bi-/uni-directional). The participants also switched their roles after trying each allocated conditions, leading to four collaborative task sessions per pair. The order of conditions was counterbalanced.

Each collaborative task was divided into two sections: symmetric and asymmetric. For the symmetric part, collaborators were

¹<https://theta360.com/en/about/theta/v.html>

²<https://www.microsoft.com/en-us/hololens>

³<https://developer.vuforia.com>

⁴<https://www.vive.com/au/product/vive-pro-eye/overview>

⁵<https://microsoft.github.io/MixedRealityToolkit-Unity/Documentation/WelcomeToMRTK.html>

⁶<https://developer.vive.com/resources/vive-sense/sdk/vive-eye-tracking-sdk-sranipal>

⁷https://valvesoftware.github.io/steamvr_unity_plugin

⁸<https://unity.com/>

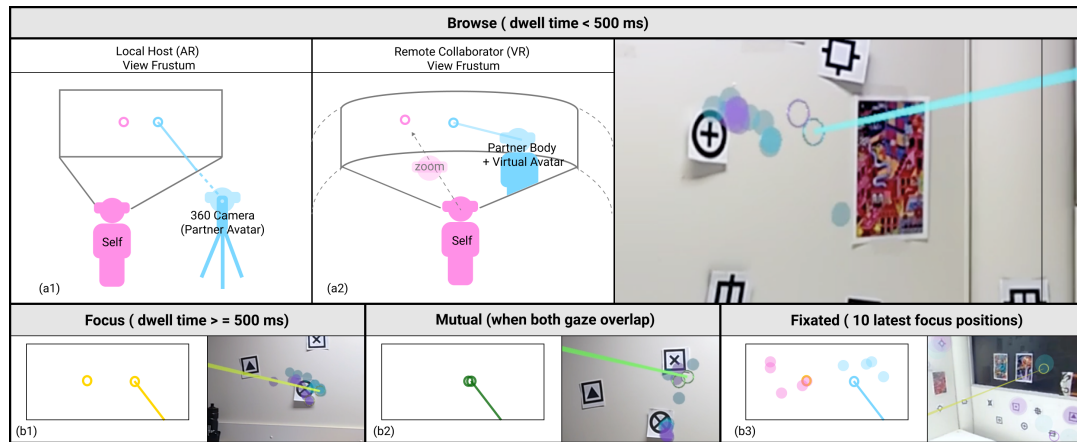


Figure 2: Visualisation of gaze cues in illustration (left) and in prototype demo (right): (a1) gaze browsing state via local host view; (a2) gaze browsing state in the remote collaborator view; (b1) gaze focus state; (b2) mutual gaze state; (b3) fixation circle-map

Table 1: Different combinations of gaze visualisation tried by each pair of participants.

	With Gaze Behaviour (All Visualisations)	No Behaviour ("Browse only")
Bi-directional (own gaze off)	Pair 1 and Pair 2	Pair 2 and Pair 3
uni-directional (own gaze off)	Pair 1 and Pair 4	Pair 3 and Pair 4

expected to work together to find one pair of duplicated symbols among three walls of thirty symbols (Fig. 1 row 1 left). For the asymmetric part, the remote collaborator would instruct the local host to locate three symbols based on the given references (Fig. 1 - row 2 middle). The symbols for the tasks were all unique and were not used repeatedly for each pair.

The study started with a brief introduction to the eyemR-Vis prototype followed by a trial use of the system in pairs prior to performing the task in each condition. The study was in a controlled lab environment: a room divided into two sections where the participants were physically blocked from seeing each other. After finishing all tasks, participants were given a questionnaire asking about their preference between the conditions as well as eight 5-point Likert scale rating questions on system usability, co-presence, mental load, and interfaces etc. (Fig. 3). Subsequently, a brief interview was held for each pair.

5.2 Results

5.2.1 Post-Study Questionnaire and Preference for the system. Figure 3 summarises the results from the questionnaire (5-point rating scale items: 1 as lowest/disagree and 5 as highest/agree). We applied One-Sample Wilcoxon Signed Rank (WSR) tests against middle value (3.0) to analyse the results and found that all participants believed the system was easy to use (Q1: $M = 4.38$, $SD = 0.52$, $p = .009$) and did what it was expected (Q2: $M = 4.38$, $SD = 0.52$, $p = .009$). The co-presence (Q3: $M = 4.13$, $SD = 0.35$, $p = .007$) and attention

Table 2: Post-Study Questionnaire

Q1: I thought the system was easy to use.
Q2: I found the system does what I expected it to do.
Q3: I felt as if I/my partner was physically present in the environment of the presentation.
Q4: My partner's focus caught my attention.
Q5: My partner's gaze visualisations are distracting.
Q6: I respond to the colour changing/colour overlay of my partners gaze.
Q7: Rate the mental effort that requires you to complete the task.
Q8: Rate the physical effort that requires you to complete the task.

cue (Q4: $M = 4.00$, $SD = 0.53$, $p = .011$) scores were positive among participants. In terms of mental (Q7: $M = 2.5$, $SD = 0.76$, $p = .102$) and physical effort (Q8: $M = 1.63$, $SD = 0.52$, $p = .009$), participants gave a relatively low rating suggesting that the system is inclined to not mentally or physically demanding. For the colour changing of gaze behaviour visualisations (Q6: $M = 3.00$, $SD = 1.09$), participants seem to have divided opinions as to which has both the lowest (1) and highest (5) rating contrary to that in preference ranking, where pairs who were comparing the condition "with gaze behaviour" vs "no behaviour" unanimously chose the former. Those could be caused by the choice of colour that needs further investigation. For the pairs who tested between "bi-directional" and "uni-directional", they all preferred the former as well. Moreover, the majority of the participants (seven out of eight) believed that the visualisations (Q5: $M = 2.38$, $SD = 0.92$) were potentially not distracting (or neutral) by giving it 3 and below rating scores.

5.2.2 Video Analysis and User Behaviours. Two researchers have performed a cross-check analysis on all videos recorded during the study. As shown in Fig. 4, we found that with gaze behavioural visualisations participants seemed to use gaze more frequently and

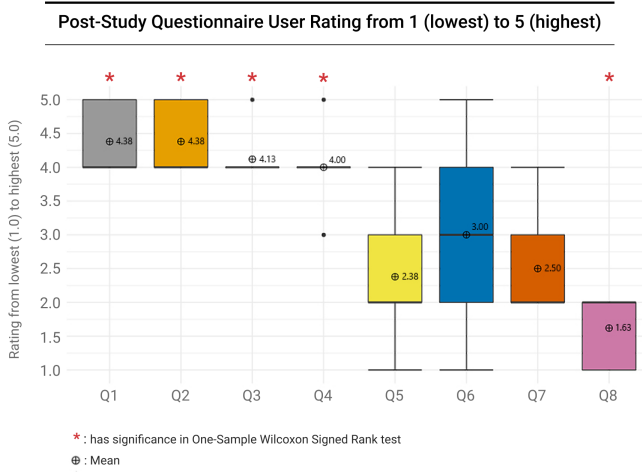


Figure 3: Questionnaire (Table 2) results on rating items: 5-point Likert scale rating from 1 (lowest/disagree) to 5 (highest/agree)

actively. The number of times the remote and local users gaze overlapping with each other (i.e. mutually gazing at the same point, a.k.a. joint gaze awareness) increased almost twice (from 54 to 93) with gaze behaviour visualisation, indicating that it appears to stimulate users to react to the change of their partner's gaze states, and thus improving gaze awareness meanwhile encouraging collaboration. Moreover, when local participants (AR user) needed to point at things during the collaboration, they used hand pointing for 11% while gaze pointing 89% of the total referential pointing counts. Both local and remote participants used the combination of gaze pointing and deictic languages ("this", "that") to refer or affirm the object of interest (e.g. "this one I am looking at") frequently. Further, participants inclined to subconsciously follow their partner's gaze during uni-directional condition because imbalanced visual feedback (i.e. seeing partner's gaze only) may have dominated the attention of the users who could not access the same visualisation of their own. This can encourage collaboration but may have negative effects on task coordination/partition. Lastly, findings from the video analysis also indicated that users have only used zoom function twice per task on average, which is not as many as we expected and it may be due to the size of the study environment being small (around 1.5m x 1.5m).

5.2.3 Interview Results and Observations. When users were asked why they preferred the bi-directional gaze over uni-directional gaze they made comments such as "so I know my partner knows where exactly I'm looking at", "I am more confident my partner knows where I refer to" and "having affirmation that we are both looking at the same object". We observed that only seeing a partner's gaze (uni-directional feedback) without their personal gaze, users may also lose the visual feedback to ensure their own gaze is successfully delivered to their partners perspective hence feeling less confident of confirmation. Participants also mentioned that behavioural visualisations are "more interactive and provide more guidance", "the

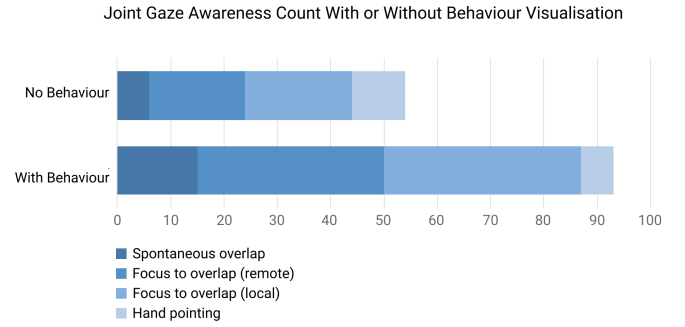


Figure 4: The count of joint gaze awareness with or without behaviour visualisations: spontaneous overlap means gaze overlap in browse state with no external stimuli spawned by the system; focus to overlap (local) means a local host focuses on an object causing the remote user to gaze at the same object; focus to overlap (remote) means a remote collaborator focuses on an object causing the local user to gaze at the same object.

feedback was more immediate and easier to notice", and it "emphasizes where someone was looking at and talking about". Half of the participants mentioned that the fixated circle-map really helped them during the task collaboration because it "made it easy to quickly identify partner's current and previous focus", and that the stacked map make it "easier to know the symbols that my partner has been looking at for a longer period of time" and they work well as a "little reminder among many symbols". Participants also mentioned that the avatar and gaze ray which simulates their partners head and gaze direction in the local user's task space enabled them to collaborate without the need for much hand gesturing or pointing although they were not physically co-located. This also aligns with our video analysis where we found local participants mostly used the combination of gaze pointing and deictic languages ("this", "that") to refer or confirm the target objects. Finally, participants felt that our gaze system accurately referred to the target objects (printed symbols on a 10cm x 10cm paper).

6 LIMITATION AND DISCUSSION

While the participants' feedback has been mostly positive, the system still has limitations. First, the sizes of AR gaze visualisations in local task space changes depending on the distance between the user's viewpoint and target surfaces in real-world (as the cursor is placed on the surface of physical objects), whereas the sizes of the same visualisations in VR 360 video remain the same. As a result, visualisations between collaborators may have a certain degree of discrepancies though participants haven't mentioned it. We plan to improve this by scaling the cursor in the AR view adaptive to the distance so that the size of it will be consistent in visual angle. Second, the Vuforia marker used for tracking the 360 camera has limited accuracy, hence the alignment between AR and VR worlds may be affected during the calibration step. Lastly, to reduce network latency due to video bandwidth the video resolution was lowered from 4k to 1080p. This was to enable the sharing of fast

eye movements to ensure synchronised visualisations. However, some of the remote participants commented on the poor video resolution, which was mentioned as a source of frustration. This could be addressed in the future by using better network bandwidth to enable sharing of high-resolution video at low latency.

7 CONCLUSION AND FUTURE WORK

In this paper, we present eyemR-Vis, a live 360 panoramic-video based Mixed Reality system that uses different visualisations of bi-directional gaze behavioural cues between a local host and a remote collaborator to enabling effective collaboration. Besides sharing 360 live video of the task space, the system also visually allows collaborators to exchange and interact with four types of gaze behavioural visualisations including "browse", "focus", "mutual", and "fixated circle-map" together with the partner's virtual avatar.

The results of an exploratory study on collaborative visual search tasks have suggested that users have found the system intuitive to use, and the visualisations of bi-directional gaze behavioural cues have made the task less physically demanding. With instant gaze visual feedback, the system has helped the participants understand each other's focus (gaze awareness) and joint focus (joint gaze awareness). According to qualitative feedback from interviews, Bi-directional gaze visualisations have also made users more confident and sure that their own gaze information is being accurately delivered.

In the future, we plan to conduct a formal experiment with multiple conditions to further explore how gaze behavioural visualisations affect pair task effectiveness, extending the results of this preliminary study. These studies will explore a number of factors, such as the change of gaze visualisation colour or shape, the representation of 3D gaze view (e.g. gaze ray, gaze frustum, or gaze arrow), and the effect of the symmetry of the task roles etc. We will continue to extend the use of different types of gaze visualisation to achieve a better user experience in MR remote collaboration for both research and practical purpose.

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